Lecture 9: Singular-Value Decomposition

Seonyeong Park, Ph.D. (✉ : sp33@illinois.edu)
Research Assistant Professor
Bioengineering
University of Illinois Urbana-Champaign
February 19, 2025

🔑 Learning Objectives

By the end of this lecture, you should be able to understand:

- Singular-value decomposition
- Measurement space and null space
- Consistency space and inconsistency space

Definition and Properties

Spectral decomposition is applied to **Hermitian opeartor**.

Singular-value decomposition (SVD) is a more general technique, applicable widely to a broader class of linear operators.

Consider a general linear opearator $\mathcal{H}$ that maps $f \in \mathbb{U}$ to $g \in \mathbb{V}$.

To analyze $\mathcal{H}$, construct a **Hermitian operator** using it:

$$\mathcal{H}^\dagger \mathcal{H} \mathbf{u}_n = \mu_n \mathbf{u}_n,$$

where

- $\mu_n = \mu_n^* \geq 0$, meaning the operator $\mathcal{H}^\dagger \mathcal{H}$ is positive semi-definite $\mathcal{H}^\dagger \mathcal{H} \succeq 0$.
- The eigenvectors $\{\mathbf{u}_n\}$ form a basis in $\mathbb{U}$.

Properties of the Basis $\{\mathbf{u}_n\}$

- Orthonormality:

$$\langle \mathbf{u}_n, \mathbf{u}_m \rangle = \mathbf{u}_n^\dagger \mathbf{u}_m = \delta_{nm} = \begin{cases} 1 & \text{if } n = m \\ 0 & \text{if } n \neq m \end{cases} \quad \text{Inner product}$$

- Completeness (Closure):

$$\sum_{n=1}^{N} \mathbf{u}_n \mathbf{u}_n^\dagger = \mathcal{I}_{\mathbb{U}} \quad \text{Outer product}$$

Use of the Positive Semi-Definite Hermitian Operators

By convention, the eigenvalues of $\mathcal{H}^\dagger\mathcal{H}$ are ordered as:

$$\mu_1 \geq \mu_2 \geq \ldots \mu_R \geq 0,$$

where

- R: Rank of $\mathcal{H}^\dagger\mathcal{H}$ ($R \leq N$, if $\mathcal{H}^\dagger\mathcal{H}$ is an $N \times N$ matrix)
- For a general matrix, R is the number of linearly independent rows or columns.
- For a Hermitian matrix, R is the number of nonzero eigenvalues.

⭐ *Note: For **general Hermitian operators**, some eigenvalues **can be negative** (* $\mu_n < 0$*), but for the form* $\mathcal{H}^\dagger\mathcal{H}$*, all* μ_n *are non-negative* ($\mu_n \geq 0$).

From the eigenvalue problem for $\mathcal{H}^\dagger\mathcal{H}$, some soluions of the corresponding problem for $\mathcal{H}\mathcal{H}^\dagger$ can be obtained:

$$\mathcal{H}^\dagger\mathcal{H}\mathbf{u}_n = \mu_n \mathbf{u}_n,$$

where the vectors $\{\mathbf{u}_n\}$ are **orthonormal**.

$$\begin{aligned} \mathcal{H}\mathcal{H}^\dagger\mathcal{H}\mathbf{u}_n &= \mathcal{H}\mu_n\mathbf{u}_n \\ \mathcal{H}\mathcal{H}^\dagger\mathcal{H}\mathbf{u}_n &= \mu_n\mathcal{H}\mathbf{u}_n \\ \mathcal{H}\mathcal{H}^\dagger\mathbf{v}_n &= \mu_n\mathbf{v}_n. \end{aligned}$$

The eigenvectors of $\mathcal{H}\mathcal{H}^\dagger$ are $\{\mathbf{v}_n\}$.

$$\begin{aligned} \langle \mathbf{v}_m, \mathbf{v}_n \rangle = \langle \mathcal{H}\mathbf{u}_m, \mathcal{H}\mathbf{u}_n \rangle = (\mathcal{H}\mathbf{u}_m)^\dagger \mathcal{H}\mathbf{u}_n = (\mathbf{u}_m)^\dagger (\mathcal{H})^\dagger \mathcal{H}\mathbf{u}_n \\ = \mathbf{u}_m^\dagger \mathcal{H}^\dagger \mathcal{H}\mathbf{u}_n = \mathbf{u}_m^\dagger \mu_n \mathbf{u}_n = \mu_n \mathbf{u}_m^\dagger \mathbf{u}_n = \mu_n \delta_{mn} \end{aligned}$$

The vectors $\{\mathbf{u}_n\}$ are **orthonormal**, and the vectors $\{\mathcal{H}\mathbf{u}_n\}$, i.e., $\{\mathbf{v}_n\}$, are **orthogonal**.

Relation Between $\{\mathbf{u}_n\}$ and $\{\mathbf{v}_n\}$

Given:

$$\mathbf{v}_n = \mathcal{H}\mathbf{u}_n$$

The $\mathbb{L}_2$ norm is computed as:

$$\|\mathbf{v}_n\| = \|\mathcal{H}\mathbf{u}_n\| = \sqrt{\langle \mathcal{H}\mathbf{u}_m, \mathcal{H}\mathbf{u}_n \rangle} = \sqrt{(\mathcal{H}\mathbf{u}_n)^\dagger \mathcal{H}\mathbf{u}_n} = \sqrt{(\mathbf{u}_n)^\dagger (\mathcal{H})^\dagger \mathcal{H}\mathbf{u}_n}$$
$$= \sqrt{\mathbf{u}_n^\dagger \mathcal{H}^\dagger \mathcal{H}\mathbf{u}_n} = \sqrt{\mathbf{u}_n^\dagger \mu_n \mathbf{u}_n} = \sqrt{\mu_n \mathbf{u}_n^\dagger \mathbf{u}_n} = \sqrt{\mu_n \delta_{nn}} = \sqrt{\mu_n}$$

If all $\mu_n \neq 0$, i.e., $\mathcal{H}\mathcal{H}^\dagger$ is **positive definite** ($\mu_n > 0$, $\mathcal{H}\mathcal{H}^\dagger \succ 0$, complete $\mathcal{H}\mathcal{H}^\dagger$), we can normalize $\{\mathbf{v}_n\}$:

$$\mathbf{v}_n = \frac{1}{\|\mathbf{v}_n\|}\mathcal{H}\mathbf{u}_n = \frac{1}{\sqrt{\mu_n}}\mathcal{H}\mathbf{u}_n$$

Thus, $\mathcal{H}^\dagger\mathcal{H}$ and $\mathcal{H}\mathcal{H}^\dagger$ have the **same eigenvalue spectrum** $\{\mu_n\}$:

$$\mathcal{H}^\dagger\mathcal{H}\mathbf{u}_n = \mu_n \mathbf{u}_n \quad \mathcal{H}\mathcal{H}^\dagger\mathbf{v}_n = \mu_n \mathbf{v}_n.$$

If some $\mu_n = 0$, i.e., $\mathcal{H}\mathcal{H}^\dagger$ is **positive semi-definite** ($\mathcal{H}\mathcal{H}^\dagger \succeq 0$, incomplete $\mathcal{H}\mathcal{H}^\dagger$), $\{\mathbf{v}_n\}$ cannot be constructed by $\mathbf{v}_n = \frac{1}{\sqrt{\mu_n}}\mathcal{H}\mathbf{u}_n$ but must be obtained by directly solving $\mathcal{H}\mathcal{H}^\dagger\mathbf{v}_n = \mu_n\mathbf{v}_n$.

Properties of the Basis $\{\mathbf{v}_n\}$

- Orthonormality:

$$\langle \mathbf{v}_n, \mathbf{v}_m \rangle = \mathbf{v}_n^\dagger \mathbf{v}_m = \delta_{nm} = \begin{cases} 1 & \text{if } n = m \\ 0 & \text{if } n \neq m \end{cases} \quad \text{Inner product}$$

- Completeness (Closure):

$$\sum_{n=1}^{N} \mathbf{v}_n \mathbf{v}_n^\dagger = \mathcal{I}_{\mathbb{V}} \quad \text{Outer product}$$

Singular-Value Decoposition (SVD)

The set $\{\mathbf{u}_n, \mathbf{v}_n, \mu_n\}$ is called the **singular system** of $\mathcal{H}$.

The **singular system** allows us to construct a representation of $\mathcal{H}$ analogous to the **spectral decomposition**:

$$\mathcal{H} = \sum_{n=1}^{R} \sqrt{\mu_n} \mathbf{v}_n \mathbf{u}_n^\dagger$$

If $\{\mathbf{u}_n\}$ forms a basis in $\mathbb{U}$, then an arbitrary vector $\mathbf{f}$ can be expressed as:

$$\mathbf{f} = \sum_{n=1}^{N} \alpha_n \mathbf{u}_n, \quad \alpha_n = \langle \mathbf{u}_n, \mathbf{f} \rangle = \mathbf{u}_n^\dagger \mathbf{f}$$

Applying $\mathcal{H}$ to $\mathbf{f}$:

$$\mathcal{H}\mathbf{f} = \sum_{n=1}^{N} \mathcal{H} \alpha_n \mathbf{u}_n = \sum_{n=1}^{N} \alpha_n \mathcal{H} \mathbf{u}_n$$

$$\mathbf{v}_n = \frac{1}{\sqrt{\mu_n}} \mathcal{H} \mathbf{u}_n, \quad \sqrt{\mu_n} \mathbf{v}_n = \mathcal{H} \mathbf{u}_n$$

$$= \sum_{n=1}^{R} \alpha_n \sqrt{\mu_n} \mathbf{v}_n$$

Applying $\mathcal{H}$ to $\mathbf{f}$:

$$\mathcal{H} = \sum_{n=1}^{R} \sqrt{\mu_n} \mathbf{v}_n \mathbf{u}_n^\dagger$$

$$\mathcal{H}\mathbf{f} = \sum_{n=1}^{R} \sqrt{\mu_n} \mathbf{v}_n \mathbf{u}_n^\dagger \sum_{m=1}^{N} \alpha_m \mathbf{u}_m = \sum_{n=1}^{R} \sum_{m=1}^{N} \alpha_m \sqrt{\mu_n} \mathbf{v}_n \mathbf{u}_n^\dagger \mathbf{u}_m$$

$$= \sum_{n=1}^{R} \sum_{m=1}^{N} \alpha_m \sqrt{\mu_n} \mathbf{v}_n \delta_{nm} = \sum_{n=1}^{R} \alpha_n \sqrt{\mu_n} \mathbf{v}_n$$

SVD and Spectral Decomposition of a Hermitian Operator

SVD of a general operator $\mathcal{H}$:

$$\mathcal{H} = \sum_{n=1}^{R} \sqrt{\mu_n} \mathbf{v}_n \mathbf{u}_n^\dagger = \mathbf{V} \mathbf{\Sigma} \mathbf{U}^\dagger,$$

where $\mathbf{\Sigma}$ is the **diagonal matrix** with the n-th diagonal element equal to $\sqrt{\mu_n}$.

$$\mathcal{H}^\dagger \mathcal{H} = \sum_{n=1}^{R} \mu_n \mathbf{u}_n \mathbf{u}_n^\dagger, \quad \mathcal{H}\mathcal{H}^\dagger = \sum_{n=1}^{R} \mu_n \mathbf{v}_n \mathbf{v}_n^\dagger$$

Both are sums of outer product operators.

Spectral decomposition of a Hermitain operator $\mathbf{A}$

$$\mathbf{A} = \sum_{n=1}^{N} \lambda_n \psi_n \psi_n^\dagger = \mathbf{\Psi \Lambda \Psi}^\dagger,$$

where $\mathbf{\Lambda}$ is the **diagonal matrix** with the n-th diagonal element equal to λ_n.

Also, a sum of outer product operators.

Division of the Domain of $\mathcal{H}$ and the Domain of $\mathcal{H}^\dagger$

Let $\mathbf{f} \in \mathbb{U}^N$ and $\mathbf{g} \in \mathbb{V}^M$. The operator $\mathcal{H}$ maps $\mathbf{f}$ to $\mathbf{g}$.

In many cases, the rank of $\mathcal{H}$ is less than the dimension of $\mathbb{U}$, i.e., $R < N$.

$$\mathbf{g} = \mathcal{H}\mathbf{f}$$

The vector $\mathbf{f}$ can be decomposed into **measuralbe** and **null** components:

$$\begin{aligned} \mathbf{g} &= \mathcal{H}\mathbf{f}_{meas} + \mathbf{0} \\ &= \mathcal{H}\mathbf{f}_{meas} + \mathcal{H}\mathbf{f}_{null} \\ &= \mathcal{H}(\mathbf{f}_{meas} + \mathbf{f}_{null}), \end{aligned}$$

where $\mathbf{f}_{null}$ is not necessarily $\mathbf{0}$, but its contribution to $\mathbf{g}$ is always zero.

The vector $\mathbf{f}$ can be expaneded, using eigenbasis $\{\mathbf{u}_n\}$:

$$\mathbf{f} = \mathbf{f}_{meas} + \mathbf{f}_{null}$$

$$\mathbf{f} = \sum_{n=1}^{N} \alpha_n \mathbf{u}_n = \sum_{n=1}^{R} \alpha_n \mathbf{u}_n + \sum_{n=R+1}^{N} \alpha_n \mathbf{u}_n$$

$$\mathbf{f}_{meas} = \sum_{n=1}^{R} \alpha_n \mathbf{u}_n, \quad \mathbf{f}_{null} = \sum_{n=R+1}^{N} \alpha_n \mathbf{u}_n$$

Null Space of $\mathcal{H}$

- $\mathbb{U}_{null} = \mathcal{N}\{\mathcal{H}\}$
- $(N-R)$-dimensional space spanned by the last $(N-R)$ eigenvectors
- Subspace of $\mathbb{U}$

If $R = N$,

- $\mathcal{N}\{\mathcal{H}\}$ is trivial. It contains only $\mathbf{f}_{null} = \mathbf{0}$.
- $\mathcal{H}$ is *one-to-one*.

Measurement Space of $\mathcal{H}$

- $\mathbb{U}_{meas} = \mathcal{N}_{\perp}\{\mathcal{H}\} = \mathcal{R}\{\mathcal{H}^{\dagger}\}$
- The orthogonal complement of the null space
- R-dimensional space spanned by the first R eigenvectors
- Subspace of $\mathbb{U}$

If $R < M$, then the data space $\mathbb{V}$ can also be divided into two nontrivial orthogonal subspaces. The data vectors $\mathbf{g}$ that are possible by $\mathcal{H}$ are in $\mathbb{V}$, but not all possible vectors in $\mathbb{V}$ can be described as $\mathcal{H}\mathbf{f}$ for some $\mathbf{f}$ in $\mathbb{U}$. The whole space $\mathbb{V}$ is not the range of $\mathcal{H}$, and $\mathcal{H}$ is not *onto*.

Consistency Space of $\mathcal{H}$

- $\mathbb{V}_{con} = \mathcal{R}\{\mathcal{H}\}$
- Range of $\mathcal{H}$
- Subspace of $\mathbb{V}$
- Example: Any ***noise-free data vector*** $\mathbf{g}$ in $\mathbb{V}_{con}$ can be described by $\mathcal{H}\mathbf{f}$ for some $\mathbf{f}$ in $\mathbb{U}$.

However, real measurements contain noise: $\mathbf{g} = \mathcal{H}\mathbf{f} + \epsilon$, which introduces inconsistency in the data.

Inconsistency Space of $\mathcal{H}$

- $\mathbb{V}_{incon} = \mathcal{R}_{\perp}\{\mathcal{H}\}$
- The orthogonal complement of $\mathbb{V}_{con}$
- Subspace of $\mathbb{V}$
- Example: The added ***noise*** ϵ belongs to $\mathbb{V}_{incon}$.

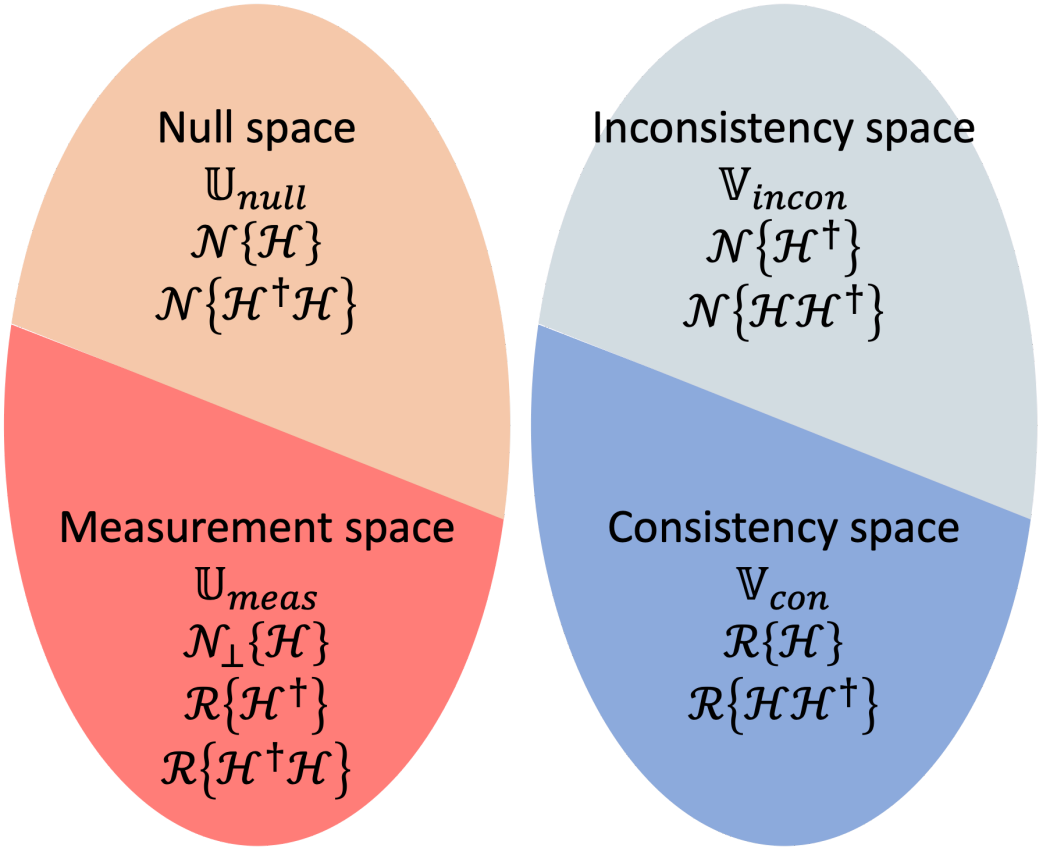
The vector $\mathbf{g}$ can be decomposed into **consistent** and **inconsistent** components:

$$\mathbf{g} = \mathbf{g}_{con} + \mathbf{g}_{incon}$$

$$\mathbf{g} = \sum_{m=1}^{M} \beta_m \mathbf{v}_m = \sum_{m=1}^{R} \beta_m \mathbf{v}_m + \sum_{m=R+1}^{M} \beta_m \mathbf{v}_m$$

$$\mathbf{g}_{con} = \sum_{m=1}^{R} \beta_m \mathbf{v}_m, \quad \mathbf{g}_{incon} = \sum_{m=R+1}^{M} \beta_m \mathbf{v}_m$$

Consider $\mathcal{H}^{\dagger}$ that maps $\mathbb{V}$ to $\mathbb{U}$. The entire $\mathbb{V}$ is the domain of $\mathcal{H}^{\dagger}$, and its range corrsponds to $\mathbb{U}_{meas}$. The null space of $\mathcal{H}^{\dagger}$ corresponds to $\mathbb{V}_{incon}$.



$\mathbb{U}$: Domain of the operator $\mathcal{H}$

$\mathbb{V}$: **Domain of the adjoint operator** $\mathcal{H}^\dagger$

$$\mathbb{U}_{meas} = \mathcal{N}_\perp\{\mathcal{H}\} = \mathcal{R}\{\mathcal{H}^\dagger\} = \mathcal{R}\{\mathcal{H}^\dagger\mathcal{H}\}$$
$$\mathbb{U}_{null} = \mathcal{N}\{\mathcal{H}\} = \mathcal{R}_\perp\{\mathcal{H}^\dagger\} = \mathcal{N}\{\mathcal{H}^\dagger\mathcal{H}\}$$
$$\mathbb{V}_{con} = \mathcal{R}\{\mathcal{H}\} = \mathcal{N}_\perp\{\mathcal{H}^\dagger\} = \mathcal{R}\{\mathcal{H}\mathcal{H}^\dagger\}$$
$$\mathbb{V}_{incon} = \mathcal{R}_\perp\{\mathcal{H}\} = \mathcal{N}\{\mathcal{H}^\dagger\} = \mathcal{N}\{\mathcal{H}\mathcal{H}^\dagger\}$$